

Google Play Store Data Preprocessing and Feature Engineering Report

1. Challenges Faced During Preprocessing

- The dataset contained missing values in several columns such as **Rating**, **Type**, **Current Ver**, and **Android Ver**.
 - Numerical columns like **Installs**, **Price**, **Size**, and **Reviews** were stored as text and required conversion to numeric format.
 - Columns contained inconsistent values such as "**Varies with device**", "**Unknown**", and "**4.4W**", which required cleaning.
 - Duplicate records were identified and removed.
 - Outliers were detected and handled using the IQR method and outlier capping.
 - Categorical variables were encoded using One-Hot Encoding.
 - Numerical features were scaled using StandardScaler and MinMaxScaler.
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2. Important Features Discovered

The following engineered features were created to improve analysis and model performance:

- **Reviews_Per_Install** – Indicates the percentage of users who left reviews after installing the app.
- **Paid_App_Flag** – Identifies whether an app is free (0) or paid (1).
- **App_Age** – Represents the number of years since the app was last updated.
- **Category_Average_Rating** – Average rating of apps within the same category.
- **Minimum_Android_Version** – Minimum Android version required to run the app.
- Date-based features such as **Year**, **Month**, **Day**, **Quarter**, and **Day Name** were extracted from the **Last Updated** column.

Feature selection using Correlation Analysis, SelectKBest, and Random Forest Feature Importance helped identify the most influential features for predicting app ratings.

3. Key Business Insights

- Apps with higher user engagement generally receive better ratings.
 - Frequently updated apps tend to maintain better ratings and user satisfaction.
 - Categories such as **Family**, **Game**, and **Tools** contain a large number of applications.
 - Free apps dominate the Google Play Store compared to paid apps.
 - The majority of apps are compatible with Android version **4.1 and above**, indicating support for a wide range of devices.
 - Reviews and installs provide valuable information about app popularity and user engagement.
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4. Final Conclusion

The Google Play Store dataset was successfully cleaned and transformed into a machine-learning-ready dataset. Missing values, duplicates, inconsistent records, and outliers were handled appropriately. Feature engineering created meaningful variables that improved the quality of the data. Feature selection techniques identified the most important variables influencing app ratings. Finally, an automated preprocessing pipeline using ColumnTransformer and Pipeline was developed to ensure consistent preprocessing while preventing data leakage. The processed dataset is now suitable for machine learning model development and business analysis.